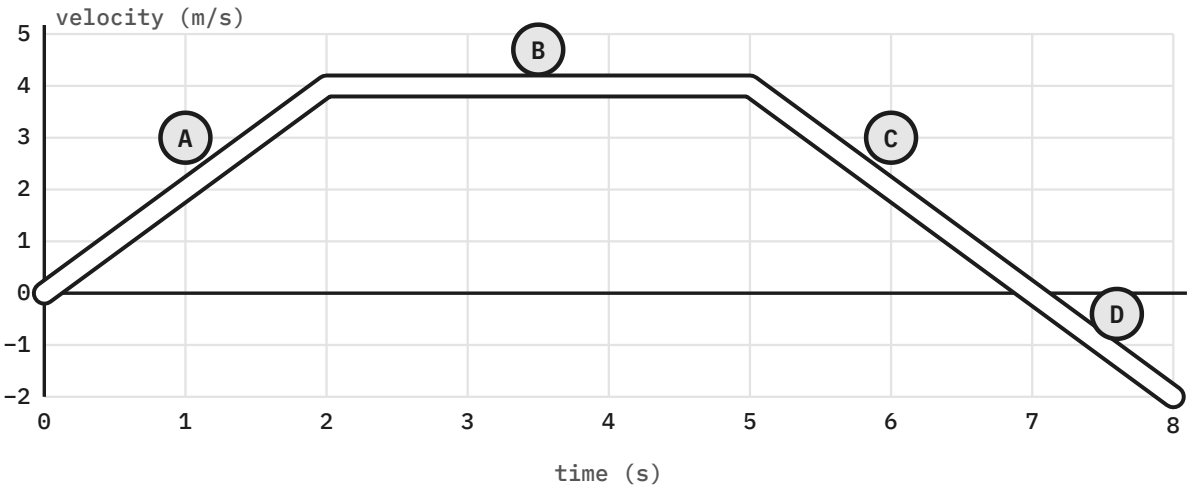
Read every part of the v–t graph

04 F

DIRECTIONS · Vertical axis = velocity. Area between the line and zero = displacement.



A · 0–2 s

speeding / steady / slowing

area = ____ m

B · 2–5 s

v = ____ m/s

area = ____ m

C · 5–7 s

v drops to ____

area = ____ m

D · 7–8 s

v ____ 0 · moving which way?

SPLIT THE ARROW

A velocity of 10 m/s points 37° above +x. v_x = 10 cos 37° = ____ m/s · v_y = 10 sin 37° = ____ m/s

TELL THE MOTION STORY

R2 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

UNIT 1 · DAY 4 · R2 · CHOICE BOARD

v–t and vectors tic-tac-toe

04 D

DIRECTIONS · Complete three in a row. Shade each square you finish.

CALCULATE v = +3 m/s for 4 s. Shade the area and find Δx.	**DRAW** Sketch a v–t graph for a bike that rides steadily, then stops.	**EXPLAIN** Why can a flat v–t line mean the object is moving?
REAL WORLD A plane flies 200 m/s at 30° above level. Find its vertical part.	**CHALLENGE ★** v = −2 m/s for 3 s, then +4 m/s for 2 s. Total Δx?	**CALCULATE** Split 20 m/s at 53° from +x into parts.
EXPLAIN Why must you say which axis an angle starts from?	**DRAW** Draw a 5 m/s arrow at 37° and its two parts.	**REAL WORLD** A boat heads east at 4 m/s; a current pushes north at 3 m/s. One-arrow speed?

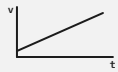
Three in a row: _____ · The square that taught me most: _____

R2 1 2 3

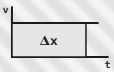
UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Say it, then write it: graphs and arrows

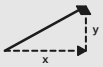
Language goal: I can **describe** a velocity–time graph.



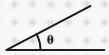
v–t graph
gráfica v–t · gráfico v–t
velocity on the up axis



area
área · área
height × width = Δx



component
componente · componente
one part of an arrow



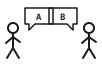
angle
ángulo · ângulo
measured from +x

English · Spanish · Portuguese. My home language: _____

1 · READ AND FILL · WORD BANK: flat · area · displacement · component · angle

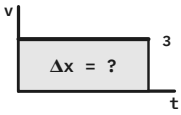
On a v–t graph, a ____ line means steady velocity. The ____ under the line is the ____. An arrow can be split into an x ____ and a y part. We measure the ____ from the +x axis.

2 · SAY IT TO A PARTNER, THEN WRITE IT



The line is flat at ____ m/s, so the object moves at a ____ velocity.
The area is ____ × ____ = ____ m, so the displacement is ____ m.

3 · SOLVE WITH THE STEPS



v = +3 m/s for 4 s on a v–t graph.

STEP 1 · HEIGHT AND WIDTH	STEP 2 · MULTIPLY	STEP 3 · SAY IT
height ____ m/s · width ____ s	____ × ____ = ____ m	"The displacement is ____ m."

4 · EXPLAIN IT · CHOOSE A FRAME

☐ A flat line does not mean stopped because ____ .
☐ The area tells me ____ because ____ .
☐ The x part is bigger than the y part because ____ .

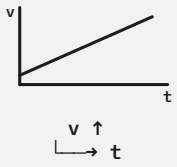
I said it ☐ · I wrote it ☐

R2 (1) (2) (3)

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Picture dictionary: graphs + vectors

Language goal: I can **describe** a graph and an arrow.



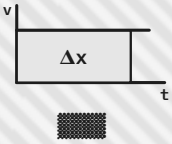
v–t graph

gráfica v–t · gráfico v–t

velocity on the up axis

my language: _____

use it: _____



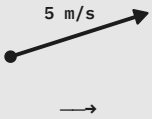
area

área · área

height × width = Δx

my language: _____

use it: _____



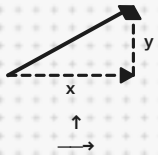
vector

vector · vetor

size + direction

my language: _____

use it: _____



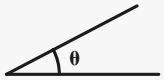
component

componente · componente

one part of an arrow

my language: _____

use it: _____



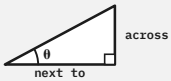
angle θ

ángulo · ângulo

measured from +x

my language: _____

use it: _____



next to
across

cos / sin

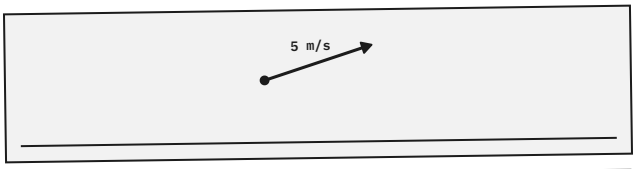
coseno/seno · cosseno/seno

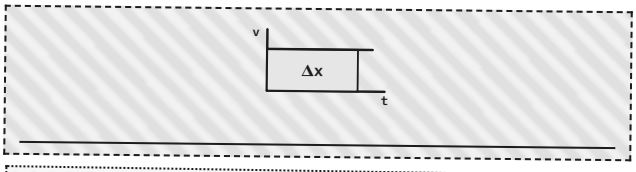
pick the right part

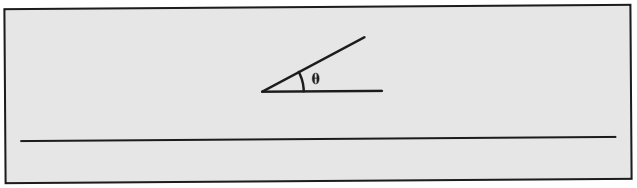
my language: _____

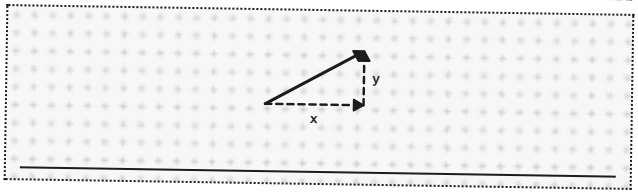
use it: _____

MATCH · WRITE THE WORD FOR EACH PICTURE









I can point and say all 6 words to a partner ☐

R2 (1) (2) (3)

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

ANTOCCI ENGINEERING · DWG U1-04 · R2

TRACKING DISH: AIM AT 2026-XJ

Story velocity: 8,500 m/s at 15° from the toward-Earth (+x) axis.

CALC 1 · TOWARD-EARTH PART

$v_x = 8{,}500 \times \cos 15° =$ ____ m/s

CALC 2 · SIDEWAYS PART

$v_y = 8{,}500 \times \sin 15° =$ ____ m/s

DRAWING · SKETCH THE ARROW AND ITS TWO PARTS (label the angle and axis)

CALC 3 · CHECK

Does $\sqrt{(v_x^2 + v_y^2)}$ give back 8,500? Show it.

CALC 4 · WHICH IS BIGGER?

Why is the sideways part small but still important?

ENGINEER'S NOTE

Write the aiming instructions for the dish crew.

R2 (1) (2) (3)

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

UNIT 1 · DAY 4 · R2 · LOOP CHAIN

I have... who has?

04 G

DIRECTIONS · Start at ★. Solve, find the tile that "has" your answer, draw an arrow, repeat.

★ START — **A**

WHO HAS…

v = +3 m/s for 4 s. What is Δx?

I have **8 m/s** — **B**

WHO HAS…

10 m/s at 37°: v_y = 10 sin 37°?

I have **stopped** — **C**

WHO HAS…

v = −2 m/s for 3 s. What is Δx?

I have **5** — **D**

WHO HAS…

The area between a v–t line and zero is…?

I have **+12 m** — **E**

WHO HAS…

A flat v–t line at 0. What is the object doing?

I have **displacement** — **F**

WHO HAS…

A flat v–t line at +5 m/s means…?

I have **−6 m** — **G**

WHO HAS…

10 m/s at 37°: v_x = 10 cos 37°?

I have **6 m/s** — **H**

WHO HAS…

Parts 3 east, 4 north. Size of the arrow?

I have **steady velocity** — **I**

END

MY PATH: A → ___ → ___ → ___ → ___ → ___ → ___ → ___ → END

R2 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else